Autonomous Industrial IoT Digital Twin & Predictive Maintenance Engine with Closed-Loop ERP Action

Capstone Project Report | Domain: Industry 4.0 | Industrial IoT | AI | SAP S/4HANA

Field	Details
Project Name	Autonomous Industrial IoT Digital Twin & Predictive Maintenance Engine with Closed-Loop ERP Action
Domain	Industry 4.0, Industrial IoT, Artificial Intelligence, SAP S/4HANA
SAP Implementation Area	SAP S/4HANA Plant Maintenance (PM) and Materials Management (MM)
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1. Project Overview

This project aims to develop an autonomous Industrial IoT Digital Twin and Predictive Maintenance Engine that monitors critical industrial equipment in real time, predicts potential failures using Artificial Intelligence, and automatically initiates maintenance and inventory actions in SAP S/4HANA.

The system will monitor a rotating industrial asset, such as a DC motor, pump, compressor, or conveyor drive, using vibration, temperature, and motor-current sensors. Sensor data will be transmitted to AWS through MQTT, processed using serverless cloud services, and analyzed using machine-learning and deep-learning models.

Unlike conventional predictive-maintenance systems that only display alerts or dashboards, this project implements a closed-loop ERP workflow. When the system detects an abnormal condition or predicts a potential failure, it will automatically create a maintenance notification or work order in SAP S/4HANA Plant Maintenance (PM). It will also check spare-part availability in SAP Materials Management (MM) and create a purchase requisition if the required part is unavailable.

The project demonstrates how IoT, cloud computing, AI, and SAP ERP can be integrated to reduce unplanned downtime, improve maintenance planning, and enable intelligent decision-making in Industry 4.0 environments.

2. Problem Statement

Industrial equipment such as motors, pumps, compressors, and conveyor systems is essential for manufacturing operations. Unexpected equipment failure can lead to production stoppages, safety risks, quality issues, high repair costs, and supply-chain disruption.

Traditional maintenance approaches have major limitations. Reactive maintenance repairs equipment only after a failure occurs, while preventive maintenance replaces parts at fixed time intervals regardless of their actual condition. This can result in unnecessary part replacement or failure to detect early signs of deterioration.

In many organizations, even when sensor data or alerts are available, maintenance teams must manually create work orders, check spare-part availability, and coordinate repairs. This manual process can delay maintenance and increase the risk of unplanned downtime.

Therefore, there is a need for an intelligent system that continuously monitors equipment condition, predicts failures before they occur, and automatically converts predictions into SAP-based maintenance and inventory actions.

3. Objectives

- To monitor vibration, temperature, and motor-current data from industrial equipment using IoT sensors.
- To transmit real-time sensor data to AWS IoT Core using MQTT.
- To process and store telemetry data using AWS Lambda, Amazon Timestream/DynamoDB, and Amazon S3.
- To create a digital twin representing the live condition and health status of the equipment.
- To apply FFT-based signal processing and feature engineering to identify abnormal vibration patterns.
- To develop an LSTM Autoencoder model for anomaly detection.
- To develop a 1D-CNN model for fault classification.
- To develop a CNN-LSTM model for Remaining Useful Life (RUL) prediction.
- To implement MLOps using DVC, MLflow, and Evidently AI.
- To build agentic AI modules for diagnosis, inventory checking, and maintenance scheduling.
- To integrate the AI engine with SAP S/4HANA through SAP BTP and OData APIs.
- To automatically create SAP PM maintenance notifications and work orders.
- To automatically check spare-part availability and create purchase requisitions through SAP MM.
- To develop an SAP Fiori dashboard for asset health, alerts, predictions, work orders, and inventory status.

4. Proposed System

The proposed system consists of eight major layers: edge sensing, cloud ingestion, data storage, digital twin, AI/ML, MLOps, agentic AI, and SAP integration.

> ***IoT Sensors → Pico W/ESP32 → AWS IoT Core → AWS Lambda → Timestream/DynamoDB + S3 → Feature Engineering → AI Models → Agentic AI → SAP BTP → SAP S/4HANA PM/MM → SAP Fiori Dashboard***

4.1 Edge and Sensing Layer

A Raspberry Pi Pico W or ESP32 microcontroller will collect data from an MPU-6050 accelerometer/gyroscope, DS18B20 temperature sensor, and ACS712 current sensor. The edge device will perform basic filtering and formatting before publishing telemetry data through MQTT.

4.2 Cloud Ingestion Layer

AWS IoT Core will receive MQTT messages from the edge device. IoT Rules will route the data to AWS Lambda functions for validation, enrichment, feature extraction, and model inference.

4.3 Data Storage Layer

Amazon Timestream or DynamoDB will store live telemetry, asset metadata, and alerts. Amazon S3 will store raw sensor data, training datasets, feature datasets, and machine-learning model artifacts.

4.4 Digital Twin and AI Layer

The digital twin will represent the current condition of the monitored asset, including sensor readings, health score, anomaly status, and predicted maintenance requirement.

The AI layer will include Isolation Forest for baseline anomaly detection, LSTM Autoencoder for unsupervised anomaly detection, 1D-CNN for fault classification, and CNN-LSTM model for RUL prediction.

FFT-based spectral analysis will convert raw vibration signals into frequency-domain features. These features will help identify patterns related to bearing wear, rotor imbalance, misalignment, overheating, and abnormal motor load.

4.5 MLOps Layer

DVC will manage dataset and model versioning. MLflow will track experiments, parameters, metrics, and model versions. Evidently AI will monitor data drift and model-performance degradation. When drift is detected, the system can trigger model retraining.

4.6 Agentic AI Layer

The agentic AI layer will include a Diagnostic Agent to interpret anomaly scores, fault classes, FFT features, and RUL values; an Inventory Agent to check spare-part availability and lead time in SAP MM; and a Scheduler Agent to determine maintenance priority and prepare SAP PM work-order data.

The agents will use retrieval-augmented generation (RAG) with maintenance manuals and standard operating procedures to provide explainable maintenance recommendations.

4.7 SAP Integration Layer

SAP BTP Integration Suite will connect the AWS-based AI engine with SAP S/4HANA using OData APIs. OAuth 2.0 will be used for secure authentication, while SAP Destination Service will securely manage connection details.

5. SAP Implementation

The core SAP implementation will be carried out in SAP S/4HANA Plant Maintenance (PM) and SAP S/4HANA Materials Management (MM).

5.1 SAP Plant Maintenance (PM)

When the AI model detects a high-risk condition, the system will create a Maintenance Notification in SAP PM. The notification will include asset details, anomaly score, predicted failure mode, RUL estimate, priority, and recommended action.

Based on business rules, the notification can be converted into a Maintenance Work Order. The work order will help maintenance planners assign technicians, schedule inspection or repair activities, and track completion status.

> ***Asset: MOTOR-001***
> ***Anomaly Score: 0.87***
> ***Predicted Failure: Bearing Wear***
> ***RUL: 38 hours***
> ***Priority: High***
> ***SAP PM Action: Create Maintenance Notification and Work Order***

5.2 SAP Materials Management (MM)

After the Diagnostic Agent identifies the required spare part, the Inventory Agent will check its availability in SAP MM. If the spare part is available, it can be planned for the maintenance work order. If it is unavailable, the system will create a Purchase Requisition in SAP MM.

> ***Required Spare Part: 6205-2RS Bearing***
> ***Available Stock: 0***
> ***Lead Time: 7 days***
> ***SAP MM Action: Create Purchase Requisition***

5.3 SAP Fiori Dashboard

The SAP Fiori dashboard will provide a single view for maintenance engineers and planners. It will display live sensor readings, asset-health score, anomaly alerts, RUL estimates, SAP work-order status, spare-part availability, and maintenance history.

6. Technology Stack

Layer	Technologies
Hardware / Edge	Raspberry Pi Pico W or ESP32, MPU-6050, DS18B20, ACS712, DC motor
Cloud	AWS IoT Core, AWS Lambda, Amazon Timestream, DynamoDB, S3, SNS, CloudWatch
Programming	Python
Data Science	NumPy, Pandas, SciPy, Scikit-learn, Matplotlib
Deep Learning	TensorFlow/Keras or PyTorch
MLOps	DVC, MLflow, Evidently AI
Agentic AI	LangGraph or CrewAI, LangChain/LlamaIndex, FAISS or Chroma
SAP	SAP BTP, Integration Suite, API Management, Destination Service, OData, S/4HANA PM, MM, Fiori

7. Expected Outcomes

- Real-time monitoring of vibration, temperature, and motor current.
- Early detection of abnormal equipment behavior.
- Prediction of Remaining Useful Life before failure.
- Automatic creation of SAP PM maintenance notifications and work orders.
- Automatic checking of spare-part availability in SAP MM.
- Automatic creation of purchase requisitions when spare parts are unavailable.
- A Fiori dashboard for asset health and maintenance visibility.
- Reduced unplanned downtime through early and automated maintenance action.
- Improved maintenance planning and spare-part readiness.

8. Conclusion

This project will demonstrate a complete Industry 4.0 predictive-maintenance solution that connects physical equipment, cloud-based AI, agentic intelligence, and SAP S/4HANA.

The major strength of the project is its closed-loop ERP action. Instead of stopping at anomaly detection, the system will convert AI-based predictions into practical SAP business actions. A sensor anomaly will lead to an AI diagnosis, RUL estimation, agentic decision-making, SAP PM work-order creation, SAP MM spare-part verification, and real-time visibility through SAP Fiori.

By integrating IoT, AWS, deep learning, MLOps, agentic AI, and SAP S/4HANA, the project will show how predictive maintenance can become an autonomous enterprise process rather than a passive analytics dashboard.